Fonds National de la Recherche Luxembourg

Thermodynamics of nonequilibrium phase transitions in driven Potts models

Massimiliano Esposito
University of Luxembourg

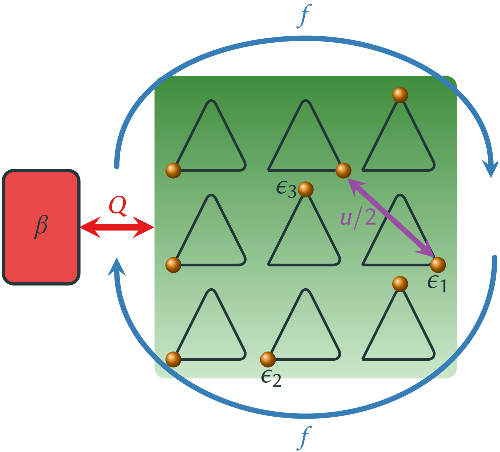
Herpich, Thingna, and Esposito, PRX **8**, 031056 (2018)
Herpich and Esposito, PRE **99**, 022135 (2019)

March 4–8, 2019

APS March Meeting

Driven Three-State Potts Model

- Open system in contact with a heat bath
- Driven by a torque f
- All-to-all interactions (pair energy $u/2$)



- Markovian master equation with Arrhenius rates satisfying local detailed balance

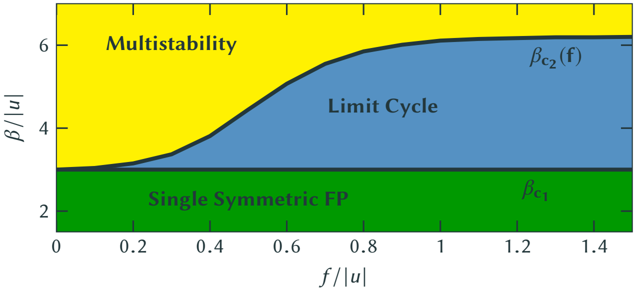
$$\dot{p}_\alpha = \sum_{\alpha'} W_{\alpha,\alpha'}\; p_{\alpha'}\,, \qquad \ln \frac{W_{\alpha\alpha'}}{W_{\alpha'\alpha}} = -\beta\left[\Delta F^{eq}(\alpha,\alpha') - \sigma(\alpha,\alpha')f\right],$$

- Mean-field dynamics given by a two-dimension nonlinear master equation

$$\dot{\overline{n}}_i(t) = \sum_{j=1}^{3} k_{ij}(\,\overline{n}_i(t), \overline{n}_j(t)\,)\,\overline{n}_j(t), \quad \ln \frac{k_{ij}}{k_{ji}} = -\beta\left[\epsilon_i - \epsilon_j + u(\overline{n}_i - \overline{n}_j) - \sigma(i,j)f\right].$$

Mean-Field Solutions

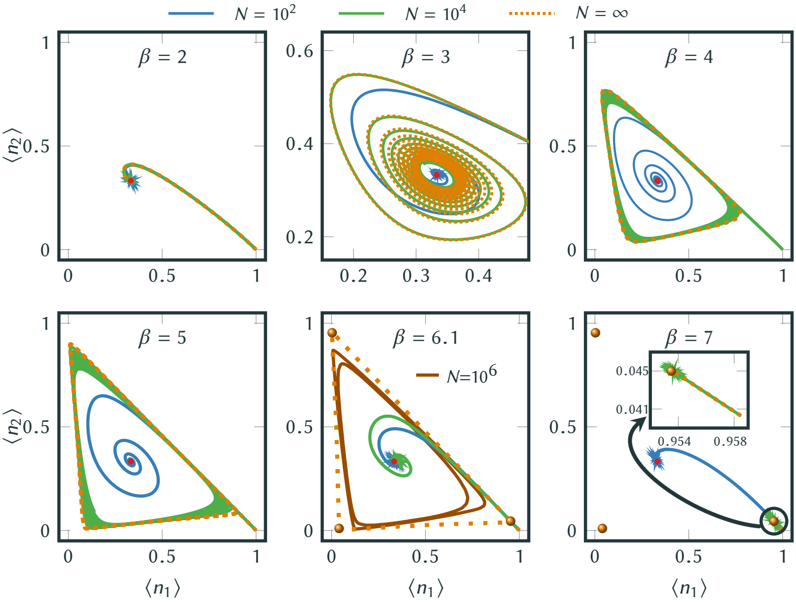
- For $\epsilon_i = \epsilon$, the nonlinear equation has a symmetric fixed point $\dot{\overline{n}}_i = \sum_{j=1}^{3} k_{ij}\, \overline{n}_j \big|_{\overline{n}_j = 1/3} = 0\,.$
- Linear stability indicates Hopf bifurcation: $\lambda_\pm = -\Gamma(\beta u + 3)\cosh\left(\frac{\beta f}{2}\right) \pm \mathrm{i}\sqrt{3}\,\Gamma \sinh\left(\frac{\beta f}{2}\right).$



Herpich, Thingna, and Esposito, PRX **8**, 031056 (2018)

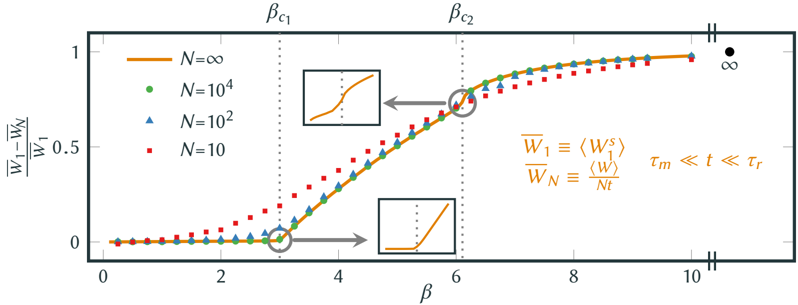
- One equilibrium (first-order) and two nonequilibrium phase transitions at $\beta = \beta_{c_{1,2}}$.
- Three distinct fixed points in the low-temperature regime ($\beta \gg \beta_c$) Energy-driven
- Symmetric solution in the high-temperature regime ($\beta \ll \beta_c$) Entropy-driven
- Thermodynamic consistency constraints stable oscillations to emerge only at intermediate temperatures ($\beta_{c_1} < \beta \le \beta_{c_2}$).

- Finite N vs. mean field:

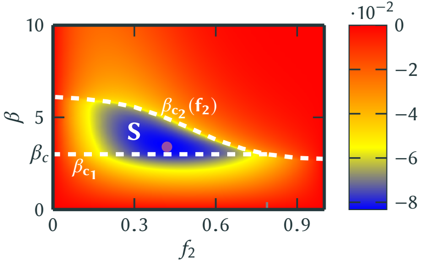


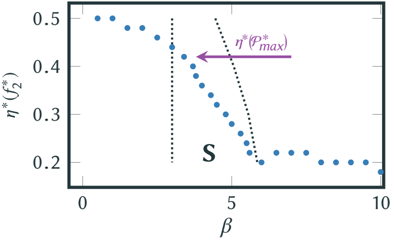
Herpich, Thingna, and Esposito, PRX **8**, 031056 (2018)

- Dissipated Work:



- Power-Efficiency Trade-off ($f = f_1 - f_2$):





Herpich, Thingna, and Esposito, PRX **8**, 031056 (2018)

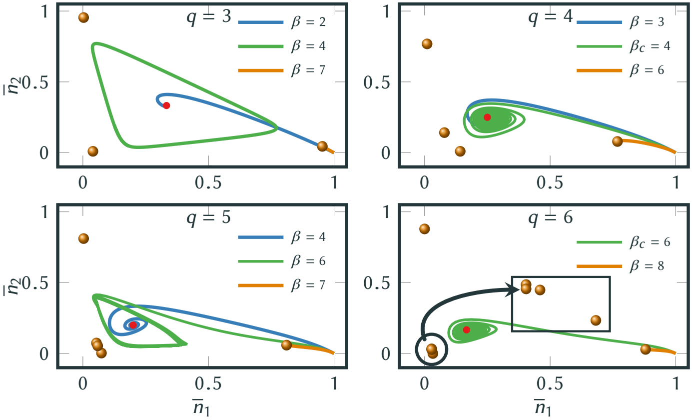
Driven q-State Potts Model

- Mean-field dynamics is governed by $(q-1)$-dimensional nonlinear Markovian master equation

$$\dot{\bar{n}}_i(t) = \sum_{j=1}^{q} k_{ij}(\bar{n}_i, \bar{n}_j)\,\bar{n}_j(t), \qquad \ln \frac{k_{ij}}{k_{ji}} = -\beta\left[\epsilon_i - \epsilon_j + u(\bar{n}_i - \bar{n}_j) - \sigma(i,j)f\right].$$

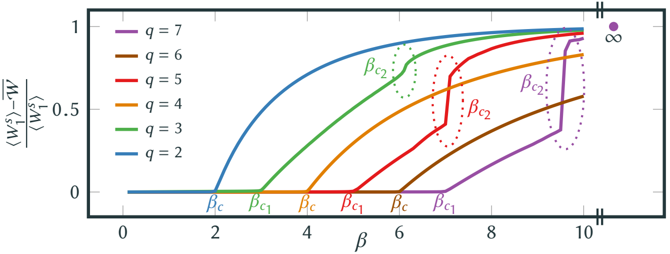
- Flat energy landscape $\epsilon_i = \epsilon\ \forall i$ and attractive interactions $u < 0$:
  - q energy-ground states $\bar{n}_i = \delta_{ij}$ in the low-temperature limit — Energy-driven
  - Symmetric solution $\bar{n}_i = 1/q$ in the high-temperature limit — Entropy-driven
  - Analytic solution of the critical Hopf-bifurcation point: $q + \beta_c u = 0$

- Stability of the limit cycle? Even vs. odd q:

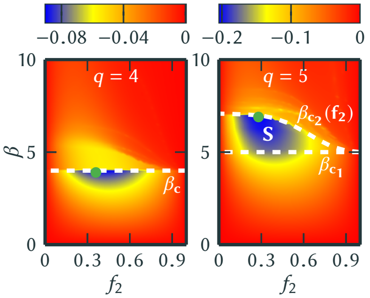


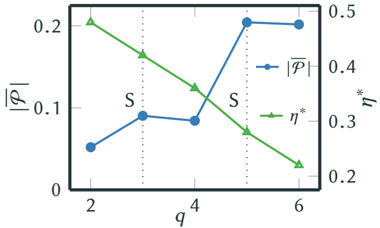
Herpich and Esposito, PRE **99**, 022135 (2019)

- Dissipated work



- Power-efficiency trade-off





Herpich and Esposito, PRE **99**, 022135 (2019)

Thank you for your attention!